AI-Enabled Coconut GenBank Intelligence and Crowdsourced WCLWD Severity Assessment

IT4010 Research Project – 2026



R26-IT-019

Comments for the Proposal

IT Number	Comment	Action
IT22904164	The proposal should clearly mention the available drone/satellite image data, define the component focus, compare it with existing GIS solutions, and strengthen the literature review and methodology justification.	verify and clearly define the use of drone/satellite images, refine the component focus as coconut planting spot detection, review related GIS-based methods, and justify the OpenCV-based approach for canopy, obstacle, and plantable area detection.
IT22236500	The meaning of “secure” in the proposal title is unclear. The proposal should clearly explain whether security refers to secure image storage, database protection, or secure classification.	Clarified security as a non-functional requirement. Security will focus on safe image upload, input validation, database protection, and controlled access. Title refined to focus on AI-based coconut germplasm classification.
IT22549204	Validity of the generated responses is questionable. Since this component depends on crowdsourced data, the proposal should clearly explain how the accuracy and trustworthiness of the collected information and generated outputs will be verified. A proper validation mechanism is needed to ensure that incorrect or misleading community responses do not affect the system.	Separate logins for CRI officers (experts) and farmers ensure only verified experts give authoritative answers. Admin verification is required for expert role approval, preventing unauthorized role changes. Confidence scoring validates responses to ensure only reliable outputs are used.
IT22088314	Validation methodology of this component is questionable. The proposal should clearly explain how the model or assessment approach will be validated, including whether cross-validation or other appropriate validation techniques will be used.	• Improved validation using an 80:20 train-test dataset split for unbiased evaluation on unseen data. • Evaluated models using Accuracy, Precision, Recall, and F1-score for comprehensive performance analysis.

The Problem





Impact of the Problem on Users and Society

Impact on Farmers

- Late disease detection reduces crop yield and income.
- Dependence on experts slows treatment decisions.

Impact on Agricultural Planning & Disease Monitoring

- Lack of accurate GIS-based land analysis.
- Delayed detection and difficulty identifying diseases like WCLWD.

Impact on Research and Gene Bank Management

- Manual germplasm identification and fragmented records.
- Inefficient data management and slow research processes.

Impact on Community & Economy

- No unified platform for crowdsourced disease reports.
- Reduced coconut production impacts the national economy.

Users & Stakeholders

Coconut Farmers

CRI Authorities

Plantation Managers

Field Officers

System Administrators

Agricultural Researchers



Data Collection Activities at Bandirippuwa

















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FACULTY OF COMPUTING

R26-IT-019

IT22904164 | Wijerathna W.M.H.N.



Land Utilization Analysis System for Coconut Genebanks

Information Technology (IT)

How the Solution Address the Sub Problem

- Developed an OpenCV-based coconut planting spot detection system.
- Analyzed satellite images to identify suitable planting areas.
- Detected obstacles such as roads, buildings, soil patches, and non-plantable regions.
- Identified plantable gaps by excluding existing coconut canopy areas.
- Generated recommended planting spots with spacing control and visual output.

User Requirements Addressed

- Allows users to upload satellite images of coconut plantation land.
- Allows users to adjust planting spot spacing based on field requirements.
- Detects coconut canopy, open gaps, and suitable plantable areas.
- Identifies obstacles such as roads, buildings, soil patches, and non-plantable regions.
- Generates a visual output showing plantable zones, obstacles, and recommended planting spots.



Contribution

- Developed the OpenCV-based coconut planting spot detection component.
- Implemented satellite image upload and validation.
- Designed canopy and vegetation detection for plantation analysis.
- Implemented obstacle detection for roads, buildings, soil patches, and non-plantable areas.
- Generated visual output with plantable zones, obstacles, and recommended planting spots.

User Feedback on Prototype

- Users understood the image upload and analysis flow clearly.
- Users suggested showing detected planting spots more visually.
- Added visual output with plantable zones, obstacles, and numbered planting spots.
- Users preferred spacing control for different field requirements.
- Added adjustable planting spacing before generating results.

IT22236500 | Randeniya A.A.S.L.B.R.P.W.R.C



Image Classification system for coconut Germplasm Identification

Information Technology (IT)



How the Solution Address the Sub Problem

- Developed an AI-based coconut germplasm classification system.
- Classified coconut images into Tall and Dwarf types for PP1.
- Trained and compared Custom CNN, MobileNetV2, and EfficientNetB0 models.
- Integrated the best model with a web prototype to show prediction, confidence score, and verification status.

User Requirements Addressed

- Allows users to upload coconut palm or nut images for germplasm identification.
- Allows users to enter field details such as Palm ID, Region, Image Type, and Notes.
- Provides AI-based prediction results for Tall and Dwarf coconut germplasm types.
- Displays the prediction confidence score to help users understand model certainty.
- Stores prediction records in the database for future research officer review and verification.



Contribution

- Prepared and preprocessed the coconut image dataset.
- Trained and evaluated Custom CNN, MobileNetV2, and EfficientNetB0.
- Selected Custom CNN as the best PP1 baseline model.
- Integrated model prediction with backend and SQLite database.
- Developed frontend for image upload, prediction result, and history view.

User Feedback on Prototype

- Users understood the upload and prediction flow clearly.
- Added Palm ID, Region, Image Type, and Notes based on feedback.
- Users preferred viewing previous records.
- Added prediction history table for saved database records.

IT22549204 | Harischandra R.D.I.D



Crowdsourced Community Intelligence Platform for Coconut Caterpillar Infestation Monitoring

Information Technology (IT)

How the Solution Address the Sub Problem

- Developed an AI-based system to detect Coconut Caterpillar Infestation (CCI) using coconut leaf images and farmer reports.
- Trained and evaluated multiple CNN models including MobileNetV2, VGG16, ResNet50, and EfficientNetB0.
- Developed a lightweight NLP model to analyze farmer text descriptions and symptoms.
- Integrated CNN and NLP models into a crowdsourcing-based web platform for CCI detection.

User Requirements Addressed

- Allows farmers to upload coconut leaf images for CCI detection.
- Supports optional symptom selection and farmer text descriptions.
- Provides fast and automated AI-based prediction results.
- Detects both CCI and Non-CCI conditions with improved accuracy.
- Supports simple and user-friendly interaction for non-technical users.
- Enables crowdsourced reporting and knowledge sharing related to coconut diseases.



Contribution

- Developed the complete CNN-based CCI image classification module.
- Implemented the hybrid NLP-based farmer text analysis system.
- Integrated CNN and NLP predictions into a unified backend system.
- Designed and developed the React frontend for farmer interaction.
- Implemented Flask APIs, SQLite database connectivity, and full-stack integration for the crowdsourcing platform.

User Feedback on Prototype

- Farmers can upload coconut leaf images for automated CCI detection.
- Farmers can optionally provide symptoms and text descriptions.
- The system provides quick AI-based prediction results.
- The platform supports accurate identification of CCI and Non-CCI conditions.
- The interface is simple and user-friendly for farmers.
- The system supports crowdsourced disease reporting and information sharing.

IT22088314 | Malshani W.A.M.S



Weligama Coconut Leaf Wilt Disease Severity Assessment Using ABCD Rule

Information Technology (IT)

How the Solution Address the Sub Problem

- Developed an AI-based system to detect WCLWD from coconut leaf images.
- Applied preprocessing, augmentation, and dataset splitting.
- Implemented CNN models (VGG16, ResNet50, InceptionV3, EfficientNet-B0) and Vision Transformer (ViT).
- Used Grad-CAM and ViT attention maps for disease localization and explainability.

User Requirements Addressed

- Fast and automated disease symptom detection.
- Improved accuracy over manual inspection.
- Explainable prediction outputs.
- Supports early symptom identification.

Contribution

- Comparative analysis of CNNs and Vision Transformer.
- Explainability using Grad-CAM and attention visualization.
- ViT showed stronger global contextual understanding.

User Feedback on Prototype

- Prototype supports faster and easier disease identification from coconut leaf images.
- Users found the prediction results simple and understandable.
- The system showed high prediction accuracy during testing.
- Backend integration successfully generated real-time disease predictions.
- Users suggested adding disease severity estimation and localization features in future improvements.

AI-Enabled Coconut GenBank Intelligence and Crowdsourced WCLWD Severity Assessment

Commercialization Potential

- Scalable AI platform for coconut disease monitoring and genebank management
- Can be adopted by CRI, agricultural institutes, and plantations
- Expandable to other crops and smart agriculture systems

Target Market & Customers

- Coconut Research Institute (CRI)
- Plantation owners & estate managers
- Farmers

Value Proposition

- GIS plantation mapping & decision support
- AI-based germplasm identification
- Real-time crowdsourced disease monitoring
- Automated WCLWD severity estimation

Cost & Revenue Model

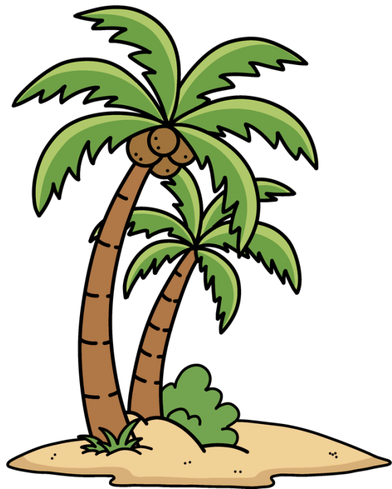
- **Investment:** AI Development, Cloud Infrastructure, Data collection
- **Recovery:** Institutional licensing, Subscriptions, Government Agricultural Programs



AI-Enabled Coconut GenBank Intelligence and Crowdsourced WCLWD Severity Assessment

GANTT CHART

TASK	OCT	NOV	DEC	JAN	FEB	MAR	MAY	JUN	JUL	AUG	SEP	OCV	DEC
1. Group Registration			20th October 2025										
2. Topic Selection													
3. TAF Documentation													
4. TAF Submission				05th January 2026									
5. Proposal Report					15th March 2026								
6. Proposal Present					16th March 2026								
7. Implementation													
8. Progress Presentation I							4th - 6th May 2026						
9. Research Paper													
10. Testing & Finalize the Project										12th - October 2026			
11. Progress Presentation II										12th - 14th October 2026			
12. Web Assessment											11th October 2026		
13. Final Presentation & Viva													1st December 2026
14. Final Report													



THANK YOU!

